NAMANGA SECONDARY SCHOOL

FORM FOUR — INFORMATION AND COMPUTER STUDIES

DATABASE PRACTICAL EXAMINATION — MARKING SCHEME

Time: 3 Hours Total: 100 Marks

GENERAL MARKING GUIDANCE

• Award marks for correct database work that satisfies the stated requirement, even where the candidate uses a different but valid Access procedure.

• For query questions, the query must both be saved with the correct name and produce the required result.

• For validation and relationship items, award the mark only when the rule actually works during testing.

• Do not award a mark for a field merely being present if the required data type, key, lookup, or validation rule is missing.

MARK DISTRIBUTION

Section	Task	Marks
A	STUDENT table	15
B	CLASS table	10
C	ACTIVITY table	10
D	Keys, lookups, validation & relationship	15
E	Four difficult queries	20
F	Advanced sorting & filtering	10
G	Forms & reports	15
H	Testing & final check	15
TOTAL		**100**

A. CREATE THE STUDENT TABLE — 15 MARKS

Item	Marks
StudentID field and Short Text type — 1 mark	1
StudentID set as Primary Key — 1 mark	1
FirstName and LastName correctly created — 2 marks	2
Gender created as Short Text — 1 mark	1
DOB created as Date/Time — 1 mark	1
ClassCode created as Short Text — 1 mark	1
Phone created as Short Text — 1 mark	1
Mark created as Number — 1 mark	1
Correct 10 student records entered — 4 marks (0.4 each)	4
Field structure and entries correctly saved — 1 mark	1

B. CREATE THE CLASS TABLE — 10 MARKS

Item	Marks
CLASS table created with ClassCode, ClassName, Stream, RoomNo, ClassTeacher — 3 marks	3
All fields correctly set as Short Text — 1 mark	1
ClassCode set as Primary Key — 1 mark	1
Correct five class records entered — 4 marks (0.8 each)	4
Table correctly saved — 1 mark	1

C. CREATE THE ACTIVITY TABLE — 10 MARKS

Item	Marks
ACTIVITY table created with all six required fields — 2 marks	2
Correct data types: ActivityID/ActivityName/Category Short Text; ActivityDate Date/Time; Participants Number; Fee Currency — 2 marks	2
ActivityID set as Primary Key — 1 mark	1
Correct eight activity records entered — 4 marks (0.5 each)	4
Table correctly saved — 1 mark	1

D. KEYS, LOOKUPS, VALIDATION AND RELATIONSHIP — 15 MARKS

Item	Marks
Gender Lookup Wizard contains only Male and Female; Limit To List = Yes — 2 marks	2
ClassCode Lookup Wizard uses CLASS.ClassCode; Limit To List = Yes — 2 marks	2
Mark validation: Between 0 And 100, with suitable validation message — 2 marks	2
Phone validation accepts exactly 10 digits beginning with 0 — 1 mark	1
Participants validation >=1 — 1 mark	1
Fee validation >=0 — 1 mark	1
ActivityDate restricted to dates in 2026 — 1 mark	1
Relationship CLASS.ClassCode → STUDENT.ClassCode created — 2 marks	2
Referential Integrity enforced — 1 mark	1
No relationship created for ACTIVITY — 1 mark	1

E. FOUR DIFFICULT QUERIES — 20 MARKS

QUERY I — Q_ICT_ACADEMIC_PERIOD (5 marks)

• Correctly uses ACTIVITY table and required output fields — 1

• Correct date criterion: 01/02/2026 to 31/03/2026 — 1

• Correct Category ICT OR Academic condition — 1

• Correct AND/OR grouping so the date condition applies to both categories — 1

• Sorts ActivityDate ascending and saves with exact name — 1

QUERY II — Q_HIGH_COLLECTION (5 marks)

• Calculated field TotalCollection = Participants * Fee — 2

• Displays required fields — 1

• Criteria TotalCollection > 50000 — 1

• Sorts TotalCollection descending and saves correctly — 1

QUERY III — Q_FORMIII_PERFORMANCE (5 marks)

• Correct join between STUDENT and CLASS — 1

• Correctly selects Form III students — 1

• PerformanceBand calculation: Excellent ≥80, Good 60–79, Needs Improvement <60 — 2

• Correct output/sort and exact query name — 1

QUERY IV — Q_CLASS_GENDER_CROSSTAB (5 marks)

• Correct join/use of CLASS and STUDENT to obtain ClassName — 1

• ClassName used as Row Heading — 1

• Gender used as Column Heading — 1

• Count of StudentID used as Value — 1

• Correct crosstab result and exact query name — 1

F. ADVANCED SORTING AND FILTERING — 10 MARKS

Item	Marks
Q_TOP5_MARKS: correct query and required fields — 1	1
Q_TOP5_MARKS: Mark sorted descending — 1	1
Q_TOP5_MARKS: Top Values set correctly to 5 — 2	2
Q_TOP5_MARKS produces correct top five marks: 91, 88, 84, 78, 76 — 1	1
Q_FORMII_LOW_OR_HIGH: correctly selects F2A or F2B — 2	2
Q_FORMII_LOW_OR_HIGH: correctly applies marks below 60 OR 80 and above, with proper grouping — 2	2
Both queries saved and run successfully — 1	1

G. FORMS AND REPORTS — 15 MARKS

Item	Marks
FORM_STUDENT based on STUDENT; Columnar layout — 2	2
FORM_CLASS based on CLASS; Tabular layout — 2	2
FORM_ACTIVITY based on ACTIVITY; Justified layout — 2	2
REPORT_STUDENT_CLASS based on Q_FORMIII_PERFORMANCE; grouped by ClassName and sorted by Mark descending — 3	3
REPORT_ACTIVITY_CATEGORY based on ACTIVITY; grouped by Category and displays ActivityName, Participants and Fee — 2	2
REPORT_PERFORMANCE displays student name, Mark and PerformanceBand, with report title and page number — 3	3
All forms/reports saved with exact names and open correctly — 1	1

H. TESTING AND FINAL CHECK — 15 MARKS

Item	Marks
Mark = 105 rejected by validation rule — 2	2
Participants = 0 rejected — 2	2
Fee = -500 rejected — 2	2
Gender = Other rejected by lookup — 2	2
Phone = 07123456 rejected — 2	2
ClassCode = F9A rejected through lookup/relationship rule — 2	2
Invalid test values removed/restored to valid records — 1	1
Final database opens correctly and four difficult queries/forms/reports work — 1	1
Database saved as NAMANGA_STUDENT_DATABASE and ready for submission — 1	1

EXPECTED QUERY RESULTS — FOR ASSESSOR REFERENCE

Query I: qualifying activities from February–March 2026 with Category ICT or Academic, correctly applying the combined date/category criteria.

Query II: TotalCollection is Participants × Fee; only records above 50,000 appear, ordered from highest to lowest.

Query III: Form III students are shown with ClassName, Stream, Mark and the correct PerformanceBand.

Query IV: Crosstab gives counts of Male and Female students for each ClassName.

Top five marks: 91, 88, 84, 78, 76.

END OF MARKING SCHEME